Hartshorne 1-27-2025

Chapter II-1

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1.2. (a) Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves on X, and $p \in X$ a point. Show that $(\ker \varphi)_p \cong \ker(\varphi_p)$ and $(\operatorname{im} \varphi)_p \cong \operatorname{im}(\varphi_p)$.
(b) Show that φ is injective (resp. surjective) if and only if φ_p is injective (resp. surjective) for all $p \in X$.
(c) Show that a sequence $\cdots \to \mathcal{F}_{i-1} \xrightarrow{\varphi_{i-1}} \mathcal{F}_i \xrightarrow{\varphi_i} \mathcal{F}_{i+1} \to \cdots$ of sheaves is exact if and only if, for all $p \in X$, the associated sequence of stalks is exact.

Proof. (a) A germ $(s, U) \in (\ker \varphi)_p \subset \mathcal{F}_p$ iff there exists $p \in V \subset U$ with $\varphi(U)(s)|_V = 0$. That is,

$$\varphi_p(s, U) = (\varphi(V)(s)|_V, V) = (0, V).$$

Similarly, a germ $(s, U) \in (\operatorname{im} \varphi)_P \subset \mathcal{G}_p$ iff there exists $p \in V \subset U$ with $s|_V = \varphi(V)(t)$. That is,

$$(s, U) = (s|_V, V) = (\varphi(V)(t), V) = \varphi_p(t, V).$$

(b) φ is injective iff $\ker \varphi \cong 0$. The zero map $\ker \varphi \to 0$ is an isomorphism iff each of its localizations $(\ker \varphi)_p \to 0$ are isomorphisms. Using (a), we see that φ is injective iff $\ker(\varphi_p) \cong (\ker \varphi)_p \cong 0$, i.e. φ_p injective, for all $p \in X$.

φ is surjective iff $\mathcal{G} = \operatorname{im} \varphi$. That is, the inclusion map $i : \operatorname{im} \varphi \to \mathcal{G}$ is an isomorphism. This occurs iff each $(\operatorname{im} \varphi)_p \to \mathcal{G}_p$ is an isomorphism, which occurs iff each $\operatorname{im}(\varphi_p) \xrightarrow{\sim} (\operatorname{im} \varphi)_p \to \mathcal{G}_p$ is an isomorphism. This map is the inclusion $\operatorname{im}(\varphi_p) \to \mathcal{G}_p$, so the result follows.

(c) Consider the following diagram:

$$\begin{tikzcd}
0 \arrow[dr] & & 0 & & 0 \arrow[dr] & & 0 & & \\
& \ker \varphi_{i-1} \arrow[ur] \arrow[dr] & & & & \ker \varphi_{i+1} \arrow[ur] \arrow[dr] & & & \\
\cdots \arrow[ur] \arrow[rr] & & \mathcal{F}_{i-1} \arrow[rr, "\varphi_{i-1}"] \arrow[dr] & & \mathcal{F}_i \arrow[ur] \arrow[rr, "\varphi_i"] & & \mathcal{F}_{i+1} \arrow[rr, "\varphi_{i+2}"] \arrow[dr] & & \cdots \\
& & & \ker \varphi_i \arrow[ur] \arrow[dr] & & & & \ker \varphi_{i+2} \arrow[ur] \arrow[dr] & \\
& & 0 \arrow[ur] & & 0 & & 0 \arrow[ur] & & 0
\end{tikzcd}$$

Clearly our original diagram is exact if and only if each diagonal $0 \to \ker \varphi_i \to \mathcal{F}_i \to \ker \varphi_{i+1} \to 0$ is exact. Thus, it suffices to show the result for short exact sequences $0 \to \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \to 0$. (b) gives us φ injective iff each φ_p is injective, ψ surjective iff each ψ_p is surjective.

If $\operatorname{im} \varphi = \ker \psi$, then this induces natural isomorphisms $\operatorname{im} \varphi_p \cong \ker \psi_p$ for all $p \in X$. Inclusions of sheaves induce inclusions of stalks, so $\operatorname{im} \varphi_p = \ker \psi_p$.

Suppose instead that $\operatorname{im}\varphi_p = \ker\psi_p$ for all $p \in X$, and fix $U \subset X$, $s \in \mathcal{F}'(U)$. For each $p \in U$, we have $\widetilde{\operatorname{im}}\,\varphi_p = \ker\psi_p$, so $\psi_p\varphi_p(s, U) = (0, U)$. That is, there is $p \subset V_p \subset U$ such that

$$\psi(U)\varphi(U)(s)|_{V_p} = 0.$$

Then the V_p form a cover of U, so the sheaf axiom gives us $\psi(U)\varphi(U)(s) = 0$, whence

$$\widetilde{\operatorname{im}}\,\varphi \subset \ker\psi.$$

Then, in fact, there is an inclusion $i : \operatorname{im}\varphi \hookrightarrow \ker\psi$ which induces the inclusions $i_p : \operatorname{im}\varphi_p \hookrightarrow \ker\psi_p$ for each $p \in V$. The latter inclusions are isomorphisms, so it follows that the former is an isomorphism as well. That is,

$$\operatorname{im}\varphi = \ker\psi.$$

$\square$

1.5. Show that a sheaf morphism $\varphi : \mathcal{F} \to \mathcal{G}$ is an isomorphism iff it is injective and surjective.

Proof. φ is an isomorphism iff each $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$ is an isomorphism. This occurs iff each φ_p is injective and surjective, which occurs iff φ is injective and surjective by the 2b. $\square$

1.8. Show that, for any $U \subset X$ open, the functor $\Gamma(U, -) : \mathrm{Sh}(X) \to \mathfrak{Ab}$ is left-exact.

Proof. Suppose $0 \to \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \to 0$ is exact. Then $\ker(\varphi(U)) = (\ker\varphi)(U) = 0$ gives us that $\varphi(U)$ is injective. It suffices to show that $\operatorname{im}(\varphi(U)) = \ker(\psi(U))$. Since $\ker\psi = \operatorname{im}\varphi = (\widetilde{\operatorname{im}}\,\varphi)^+$, it suffices to show that $\widetilde{\operatorname{im}}\,\varphi$ is a sheaf. $\varphi(U)$ is injective, so $\varphi(U) : \mathcal{F}'(U) \to \widetilde{\operatorname{im}}\,\varphi(U)$ is an isomorphism of abelian groups. Then $\varphi : \mathcal{F}' \to \widetilde{\operatorname{im}}\,\varphi$ is an isomorphism of presheaves, so $\widetilde{\operatorname{im}}\,\varphi$ is a sheaf. $\square$

1.16. *Flasque Sheaves.* A sheaf $\mathcal{F}$ on a topological space X is *flasque* if for every inclusion $V \subset U$ of open sets, the restriction map $\mathcal{F}(U) \to \mathcal{F}(V)$ is surjective.

(a) Show that a constant sheaf on an irreducible topological space is flasque.

(b) If $0 \to \mathcal{F}' \to \mathcal{F} \xrightarrow{\varphi} \mathcal{F}'' \to 0$ is an exact sequence of sheaves, and if $\mathcal{F}'$ is flasque, then for any open set U, the sequence $0 \to \mathcal{F}'(U) \to \mathcal{F}(U) \xrightarrow{\varphi} \mathcal{F}''(U) \to 0$ of abelian groups is also exact.

(c) If $0 \to \mathcal{F}' \to \mathcal{F} \xrightarrow{\varphi} \mathcal{F}'' \to 0$ is an exact sequence of sheaves, and if $\mathcal{F}'$ and $\mathcal{F}$ are flasque, then $\mathcal{F}''$ is flasque.

(d) If $f : X \to Y$ is a continuous map and $\mathcal{F}$ is a flasque sheaf on X, then $f_*\mathcal{F}$ is a flasque sheaf on Y.

(e) Let $\mathcal{F}$ be any sheaf on X. We define a new sheaf $\mathcal{G}$, called the sheaf of *discontinuous sections* of $\mathcal{F}$ as follows. For each open set $U \subset X$, $\mathcal{G}(U)$ is the set of maps $s : U \to \bigcup_{p \in U} \mathcal{F}_p$ such that for each $p \in U$, $s(p) \in \mathcal{F}_p$. Show that $\mathcal{G}$ is a flasque sheaf, and that there is a natural injective morphism $\mathcal{F} \rightarrowtail \mathcal{G}$.

Proof. (a) Let X be irreducible, $\overline{A}$ a constant sheaf on X. Any open subset of an irreducible topological space is connected, so $\overline{A}(U) \cong A$ for all open sets $U \subset X$. Then the restriction map $\overline{A}(U) \to \overline{A}(V)$ is the identity $A \to A$, which is in particular surjective.

(b) The sequence is left exact by (Ex. 1.8), so it suffices to show surjectivity of $\varphi(U) : \mathcal{F}(U) \to \mathcal{F}''(U)$. First, take $\varphi(s_1) \in \mathcal{F}''(U_1)$ and $\varphi(s_2) \in \mathcal{F}''(U_2)$ such that $\varphi(s_1)|_{U_{12}} = \varphi(s_2)|_{U_{12}}$. Then $s_1 - s_2 \in \mathcal{F}'(U_{12})$, so $\mathcal{F}'$ flasque supplies us with $t_{12} \in \mathcal{F}'(U_1 \cup U_2)$ such that

$t_{12}|_{U_{12}} = s_1 - s_2$. Then $s_1 \in \mathcal{F}(U_1)$ and $t_{12} + s_2 \in \mathcal{F}(U_2)$ glue to a section $s \in \mathcal{F}(U_1 \cup U_2)$ such that

$$s|_{U_1} = s_1, \qquad \varphi(s)|_{U_2} = \varphi(t_{12} + s_2) = \varphi(s_2).$$

Fix $s'' \in \mathcal{F}''(U)$. By (Ex. 1.4), we can take $s_i \in \mathcal{F}(U_i)$ such that $\varphi(s_i) = s''|_{U_i}$. Consider the poset

$$\Sigma = \{(s \in \mathcal{F}(V), V \subset U) \ : \ \varphi(s) = s''|_V\},$$

ordered by

$$(s, V) \leqslant (t, W) \iff V \subset W \ \&\ s = t|_V.$$

It is clearly nonempty, as each $(s_i, U_i) \in \Sigma$. If $C = \{(t_j, W_j)\}_{j \in J}$ is a nonempty chain, then we can glue to receive a section $t \in \mathcal{F}(\bigcup_{j \in J} W_j)$ such that $\varphi(t)|_{W_j} = \varphi(t_j) = s''|_{W_j}$ for all $j \in J$. It follows by uniqueness of gluing in $\mathcal{F}''$ that $\varphi(t) = s''|_{\bigcup_{j \in J} W_j}$, whence $(t, \bigcup_{j \in J} W_j) \in \Sigma$.

By the proceeding paragraph Zorn's lemma supplies us with a maximal element $(s, V) \in \Sigma$. Suppose, by way of contradiction, that $V \neq U$. We can take some U_i such that $V \subsetneq V \cup U_i$. The construction from the first paragraph supplies us with a section $t \in \mathcal{F}(V \cup U_i)$ such that $t|_V = s$, and $\varphi(t)|_{U_i} = \varphi(s_i)$. Then uniqness of gluing in $\mathcal{F}''$ gives us $\varphi(t) = s''|_{V \cup U_i}$, so that $(s, V) < (t, V \cup U_i) \in \Sigma$, contradicting maximality of (s, V). Hence, $V = U$, so we have $s \in \mathcal{F}(U)$ such that $\varphi(s) = s''$, as desired.

(c) The following diagram is exact by (b), whence a trivial diagram chase gives us our result.

$$\begin{tikzcd}
\mathcal{F}(U) \arrow[r, "\varphi(U)"] \arrow[d] & \mathcal{F}''(U) \arrow[d] & \\
\mathcal{F}(V) \arrow[r, "\varphi(V)"] \arrow[d] & \mathcal{F}''(V) \arrow[r] & 0 \\
0 & &
\end{tikzcd}$$

(d) Fix open subsets $V \subset U \subset Y$. Then the restriction $f_*\mathcal{F}(U) \to f_*\mathcal{F}(V)$ is simply the restriction

$$\mathcal{F}(f^{-1}(U)) \to \mathcal{F}(f^{-1}(V)),$$

which is surjective since $\mathcal{F}$ is flasque.

(e) Trivial. $\square$

1.20. *Subsheaf with Supports.* Let Z be a closed subset of X, and let $\mathcal{F}$ be a sheaf on X. We define $\Gamma_Z(X, \mathcal{F})$ to be the subgroup of $\Gamma(X, \mathcal{F})$ consisting of all sections whose support (Ex. 1.14) is contained in Z.

(a) Show that the presheaf $V \mapsto \Gamma_{Z \cap V}(V, \mathcal{F}|_V)$ is a sheaf. It is called the subsheaf of $\mathcal{F}$ with supports in Z and is denoted by $\mathcal{H}^0_Z(\mathcal{F})$.

(b) Let $U = X - Z$, and let $j : U \to X$ be the inclusion. Show there is an exact sequence of sheaves on X

$$0 \to \mathcal{H}^0_Z(\mathcal{F}) \to \mathcal{F} \to j_*(\mathcal{F}|_U).$$

Furthermore, if $\mathcal{F}$ is flasque, the map $\mathcal{F} \to j_*(\mathcal{F}|_U)$ is surjective.

Proof. (a) It suffices to show that the gluing $s \in \Gamma(V, \mathcal{F}|_V)$ of sections $s_i \in \Gamma(V_i, \mathcal{F}|_V)$ with $\operatorname{Supp} s_i \subset V_i \cap Z$ must have $\operatorname{Supp} s \subset V \cap Z$. Then $s_x = (s_i)_x$ for all $x \in U_i$ gives us

$$\operatorname{Supp} s = \bigcup_i \operatorname{Supp} s_i \subset \bigcup_i V_i \cap Z = V \cap Z,$$

as desired.

(b) Note that

$$\rho(V) = \rho^V_{V\cap U} : \mathcal{F}(V) \to \mathcal{F}(V \cap U) = \mathcal{F}|_U(j^{-1}(V)) = j_*(\mathcal{F}|_U),$$

so ρ is indeed surjective if $\mathcal{F}$ is flasque. Now, let $\mathcal{F}$ be arbitrary, take $x \in X$, and consider

$$0 \to \mathcal{H}^0_Z(\mathcal{F})_x \hookrightarrow \mathcal{F}_x \xrightarrow{\rho} j_*(\mathcal{F}|_U)_x.$$

Suppose first that $x \in U$. Then $\mathcal{H}^0_Z(\mathcal{F}) = 0$, so it suffices to show that $\mathcal{F}_x = j_*(\mathcal{F}|_U)_x$. We have that

$$\begin{aligned} \rho_x &= \varinjlim_{V \ni x} \rho^V_{V\cap U} : \mathcal{F}(V) \to \mathcal{F}(V \cap U) \\ &= \varinjlim_{U \supset V \ni x} \mathrm{id}_{\mathcal{F}}(V) : \mathcal{F}(V) \to \mathcal{F}(V) \end{aligned}$$

is a direct limit of isomorphisms, and hence an isomorphism.

Now, take $x \in Z$, and let $(s, V) \in \mathcal{F}_x$. We have $\rho_x(s, V) = 0$ iff there exists some open set $x \in W \subset V$ with $s|_{W\setminus Z} = 0$. If $(s, V) \in \mathcal{H}^0_Z(\mathcal{F})_x$, then we can take $x \in W \subset V$ such that $s|_W \in \mathcal{H}^0_Z(\mathcal{F}, W)$. Then $\operatorname{Supp} s|_W \subset Z$ implies $s|_{W\setminus Z} = 0$, as desired. Conversely, if $s|_{W\setminus Z} = 0$, then $s|_W \in \mathcal{H}^0_Z(\mathcal{F}, W)$ gives us $(s, V) = (s|_W, W) \in \mathcal{H}^0_Z(\mathcal{F})_x$. We conclude that the sequence is, indeed, exact. $\square$